Task 9 – Experimental performance

The following measurements were generated by running the supplied script in the current execution environment. They are illustrative and environment-specific; rerun the script on your own laptop and use its CSV values for the final submission. Search timing excludes one-time sorting/table construction; the preprocessing costs are discussed separately.

Records (n)	Algorithm	Mean time (ms)	Median time (ms)	Mean comparisons/lookups	Preprocessing
1,000	Linear Search	0.058838	0.017631	479.30	No sorting required
1,000	Recursive Binary Search	0.001911	0.001665	9.00	Sort once: O(n log n)
1,000	Hash Table Search	0.000417	0.000281	1.00	Build hash table once: expected O(n)
10,000	Linear Search	0.897288	0.487795	4,896.93	No sorting required
10,000	Recursive Binary	0.071547	0.002521	12.20	Sort once: O(n log n)

	Search				
10,000	Hash Table Search	0.026078	0.000374	1.00	Build hash table once: expected O(n)
100,000	Linear Search	12.441707	11.321074	51,220.87	No sorting required
100,000	Recursive Binary Search	0.021422	0.004004	15.33	Sort once: O(n log n)
100,000	Hash Table Search	0.001114	0.001119	1.00	Build hash table once: expected O(n)
1,000,000	Linear Search	244.479962	210.173546	521,474.03	No sorting required
1,000,000	Recursive Binary	0.009433	0.006982	18.93	Sort once: O(n log n)
	Search				
1,000,000	Hash Table Search	0.008276	0.001472	1.00	Build hash table once: expected O(n)